

Experiment (8)

Virtual Private Network (VPN)

8.1 Objectives

To achieve VPN Usage

8.2 Equipment / Program Requirements:

Cisco Packet Tracker

eNSP

8.3 Experiment Setup:

- 1- make topology as shown
- 2- then make configuration
- 3- check vpn
- 4- ping to test

8.4 Simulation Results:

eNSP:

R4:



```
[Huawei]undo ip route
[Huawei]undo ip route-static 12.0.0.0 24 g0/0/0
[Huawei]ip route-static 12.0.0.0 24 11.0.0.2
[Huawei] User interface con0 is available

Please Press ENTER.

<Huawei>
<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]acl nu
[Huawei]acl number 3002
[Huawei-acl-adv-3002]rule 10 permit ip source 10.0.0.0 0.0.0.255 des
[Huawei-acl-adv-3002]rule 10 permit ip source 10.0.0.0 0.0.0.255 destination 12.
0.0.0 0.0.0.255
[Huawei-acl-adv-3002]q
[Huawei]ipse
[Huawei]ipsec proposal pl
[Huawei-ipsec-proposal-pl]esp
[Huawei-ipsec-proposal-pl]esp a
[Huawei-ipsec-proposal-pl]esp authentication-algorithm sha1
[Huawei-ipsec-proposal-pl]esp e
[Huawei-ipsec-proposal-pl]esp encryption-algorithm des
[Huawei-ipsec-proposal-pl]q
[Huawei]ip se
[Huawei]ips
[Huawei]ipsec pol
[Huawei]ipsec policy pl 1 man
[Huawei]ipsec policy pl 1 manual
[Huawei-ipsec-policy-manual-pl-1]tunne
[Huawei-ipsec-policy-manual-pl-1]tu
[Huawei-ipsec-policy-manual-pl-1]tunnel loa
[Huawei-ipsec-policy-manual-pl-1]tunnel loc
[Huawei-ipsec-policy-manual-pl-1]tunnel local 11.0.0.1
[Huawei-ipsec-policy-manual-pl-1]tunn
[Huawei-ipsec-policy-manual-pl-1]tunnel remote 11.0.0.2
[Huawei-ipsec-policy-manual-pl-1]secu
[Huawei-ipsec-policy-manual-pl-1]security acl 3002
```

```

[Huawei]ips
[Huawei]ipsec pol
[Huawei]ipsec policy pl 1 man
[Huawei]ipsec policy pl 1 manual
[Huawei-ipsec-policy-manual-pl-1]tunne
[Huawei-ipsec-policy-manual-pl-1]tu
[Huawei-ipsec-policy-manual-pl-1]tunnel loa
[Huawei-ipsec-policy-manual-pl-1]tunnel loc
[Huawei-ipsec-policy-manual-pl-1]tunnel local 11.0.0.1
[Huawei-ipsec-policy-manual-pl-1]tunn
[Huawei-ipsec-policy-manual-pl-1]tunnel remote 11.0.0.2
[Huawei-ipsec-policy-manual-pl-1]secu
[Huawei-ipsec-policy-manual-pl-1]security acl 3002
[Huawei-ipsec-policy-manual-pl-1]prop
[Huawei-ipsec-policy-manual-pl-1]proposal pl
[Huawei-ipsec-policy-manual-pl-1]sa sp
[Huawei-ipsec-policy-manual-pl-1]sa spi in
[Huawei-ipsec-policy-manual-pl-1]sa spi inbound es
[Huawei-ipsec-policy-manual-pl-1]sa spi inbound esp 12345
[Huawei-ipsec-policy-manual-pl-1]sa spi outbound esp 54321
[Huawei-ipsec-policy-manual-pl-1]sa str
[Huawei-ipsec-policy-manual-pl-1]sa string-key inb
[Huawei-ipsec-policy-manual-pl-1]sa string-key inbound es
[Huawei-ipsec-policy-manual-pl-1]sa string-key inbound esp s
[Huawei-ipsec-policy-manual-pl-1]sa string-key inbound esp simple sha
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp sss
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]ip sec
[Huawei]ips
[Huawei]ipsec policy pl 1 man
[Huawei]ipsec policy pl 1 manual
[Huawei-ipsec-policy-manual-pl-1]sa st
[Huawei-ipsec-policy-manual-pl-1]sa string-key ou
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp sha
^
Error: Wrong parameter found at '^' position.
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp sim
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp simple sha
[Huawei-ipsec-policy-manual-pl-1]q
[Huawei]int g0/0/0
[Huawei-GigabitEthernet0/0/0]ipse
[Huawei-GigabitEthernet0/0/0]ipsec policy pl
[Huawei-GigabitEthernet0/0/0]

```

R4

```
[Huawei-ipsec-policy-manual-pl-1]sa string-key inbound esp s
[Huawei-ipsec-policy-manual-pl-1]sa string-key inbound esp simple sha
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp sss
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]ip sec
[Huawei]ips
[Huawei]ipsec policy pl 1 man
[Huawei]ipsec policy pl 1 manual
[Huawei-ipsec-policy-manual-pl-1]sa st
[Huawei-ipsec-policy-manual-pl-1]sa string-key ou
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp sha
^
```

Error: Wrong parameter found at '^' position.

```
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp sim
[Huawei-ipsec-policy-manual-pl-1]sa string-key outbound esp simple sha
[Huawei-ipsec-policy-manual-pl-1]q
[Huawei]int g0/0/0
[Huawei-GigabitEthernet0/0/0]ipse
[Huawei-GigabitEthernet0/0/0]ipsec policy pl
[Huawei-GigabitEthernet0/0/0]display ip rou
[Huawei-GigabitEthernet0/0/0]display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destinations : 7 Routes : 7

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/24	Direct	0	0	D	10.0.0.1	GigabitEthernet
0/0/1						
10.0.0.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
11.0.0.0/24	Direct	0	0	D	11.0.0.1	GigabitEthernet
0/0/0						
11.0.0.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
12.0.0.0/24	Static	60	0	RD	11.0.0.2	GigabitEthernet
0/0/0						
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

```
[Huawei-GigabitEthernet0/0/0]
```

R5:

```
[Huawei-ipsec-policy-manual-p2-2]sa string-key out
[Huawei-ipsec-policy-manual-p2-2]sa string-key outbound esp si
[Huawei-ipsec-policy-manual-p2-2]sa string-key outbound esp simple sha
[Huawei-ipsec-policy-manual-p2-2]q
[Huawei]int g0/0/0
[Huawei-GigabitEthernet0/0/0]ipse
[Huawei-GigabitEthernet0/0/0]ipsec poli
[Huawei-GigabitEthernet0/0/0]ipsec policy p2
[Huawei-GigabitEthernet0/0/0] User interface con0 is available
```

Please Press ENTER.

```
<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]display ip rou
[Huawei]display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destinations : 7 Routes : 7

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0/0/0 10.0.0.0/24	Static	60	0	RD	11.0.0.1	GigabitEthernet
0/0/0 11.0.0.0/24	Direct	0	0	D	11.0.0.2	GigabitEthernet
0/0/0 11.0.0.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0 12.0.0.0/24	Direct	0	0	D	12.0.0.1	GigabitEthernet
0/0/1 12.0.0.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1 127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[Huawei]

```
<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]acl num
[Huawei]acl number 3001
[Huawei-acl-adv-3001]rule 5 permit ip so
[Huawei-acl-adv-3001]rule 5 permit ip source 12.0.0.0 0.0.0.255 des
[Huawei-acl-adv-3001]rule 5 permit ip source 12.0.0.0 0.0.0.255 destination 10.
0.0.0 0.0.0.255
^
Error: Wrong parameter found at '^' position.
[Huawei-acl-adv-3001]rule 5 permit ip source 12.0.0.0 0.0.0.255 destination 10.0.
0.0.0 0.0.0.255
^
Error: Wrong parameter found at '^' position.
[Huawei-acl-adv-3001]rule 5 permit ip source 12.0.0.0 0.0.0.255 destination 10
0.0.0 0.0.0.255
[Huawei-acl-adv-3001]q
[Huawei]ipse
[Huawei]ipsec propo
[Huawei]ipsec proposal p2
[Huawei-ipsec-proposal-p2]esp a
[Huawei-ipsec-proposal-p2]esp authentication-algorithm shal
[Huawei-ipsec-proposal-p2]esp en
[Huawei-ipsec-proposal-p2]esp encryption-algorithm des
[Huawei-ipsec-proposal-p2]q
[Huawei]ipse
[Huawei]ipsec polll
[Huawei]ipsec pol
[Huawei]ipsec policy p2 2 man
[Huawei]ipsec policy p2 2 manual
[Huawei-ipsec-policy-manual-p2-2]secu
[Huawei-ipsec-policy-manual-p2-2]security acl 3001
[Huawei-ipsec-policy-manual-p2-2]proposal p2
[Huawei-ipsec-policy-manual-p2-2]tunnel
[Huawei-ipsec-policy-manual-p2-2]tunnel lo
[Huawei-ipsec-policy-manual-p2-2]tunnel local 11.0.0.2
[Huawei-ipsec-policy-manual-p2-2]tunnel remote 11.0.0.1
[Huawei-ipsec-policy-manual-p2-2]sa spi
[Huawei-ipsec-policy-manual-p2-2]sa spi in
```

```
[Huawei-ipsec-policy-manual-p2-2]sa spi inbound esp
[Huawei-ipsec-policy-manual-p2-2]sa spi inbound esp ?
  INTEGER<256-4294967295> The value of security parameter index(SPI)

[Huawei-ipsec-policy-manual-p2-2]sa spi inbound esp 54321
[Huawei-ipsec-policy-manual-p2-2]sa spi
[Huawei-ipsec-policy-manual-p2-2]sa spi ou
[Huawei-ipsec-policy-manual-p2-2]sa spi outbound es
[Huawei-ipsec-policy-manual-p2-2]sa spi outbound esp 12345
[Huawei-ipsec-policy-manual-p2-2]sa s
[Huawei-ipsec-policy-manual-p2-2]sa str
[Huawei-ipsec-policy-manual-p2-2]sa string-key in
[Huawei-ipsec-policy-manual-p2-2]sa string-key inbound esp si
[Huawei-ipsec-policy-manual-p2-2]sa string-key inbound esp simple sha
[Huawei-ipsec-policy-manual-p2-2]sa string-key out
[Huawei-ipsec-policy-manual-p2-2]sa string-key outbound esp si
[Huawei-ipsec-policy-manual-p2-2]sa string-key outbound esp simple sha
[Huawei-ipsec-policy-manual-p2-2]q
[Huawei]int g0/0/0
[Huawei-GigabitEthernet0/0/0]ipse
[Huawei-GigabitEthernet0/0/0]ipsec poli
[Huawei-GigabitEthernet0/0/0]ipsec policy p2
[Huawei-GigabitEthernet0/0/0] User interface con0 is available
```

Please Press ENTER.

```
<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]display ip rou
[Huawei]display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destinations : 7 Routes : 7

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/24	Static	60	0	RD	11.0.0.1	GigabitEthernet

```
PC5
Basic Config Command MCPacket UdpPacket Console
From 12.0.0.2: bytes=32 seq=2 ttl=126 time=141 ms
From 12.0.0.2: bytes=32 seq=3 ttl=126 time=93 ms
From 12.0.0.2: bytes=32 seq=4 ttl=126 time=94 ms
From 12.0.0.2: bytes=32 seq=5 ttl=126 time=78 ms

--- 12.0.0.2 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/101/141 ms

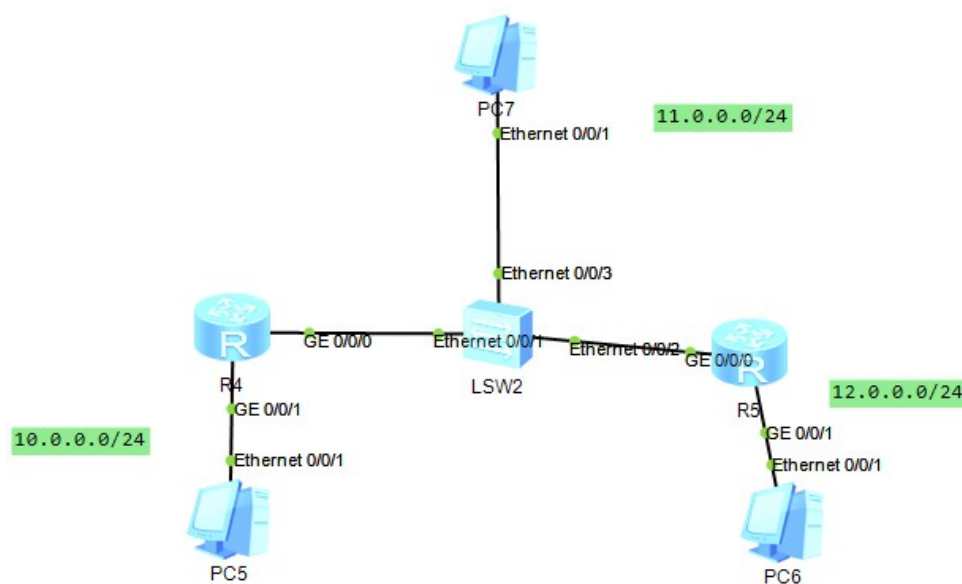
PC>ping 12.0.0.2

Ping 12.0.0.2: 32 data bytes, Press Ctrl_C to break
From 12.0.0.2: bytes=32 seq=1 ttl=126 time=78 ms
From 12.0.0.2: bytes=32 seq=2 ttl=126 time=94 ms
From 12.0.0.2: bytes=32 seq=3 ttl=126 time=110 ms
From 12.0.0.2: bytes=32 seq=4 ttl=126 time=93 ms
From 12.0.0.2: bytes=32 seq=5 ttl=126 time=94 ms

--- 12.0.0.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 78/93/110 ms

PC>
```

Wire shark doesn't work with me



Cisco Packet Tracker:

R6:

 Router6

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0/
^
% Invalid input detected at '^' marker.

Router(config)#int g0/0
Router(config-if)#ip add
Router(config-if)#ip address
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int se0/3/0
Router(config-if)#ip add
Router(config-if)#ip address 11.0.0.1 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#
```

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☐ Top

IOS Command Line Interface

```
Router(config)#int tunnel 0
Router(config-if)#ip add
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#tunnel source 11.0.0.1
^
% Invalid input detected at '^' marker.

Router(config-if)#tunnel source se0/3/0
Router(config-if)#tunnel des
Router(config-if)#tunnel destination 12.0.0.2
Router(config-if)#
01:25:07: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on Tunnel0 from LOADING to FULL,
Loading Done

Router(config-if)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 11.0.0.2 to network 0.0.0.0

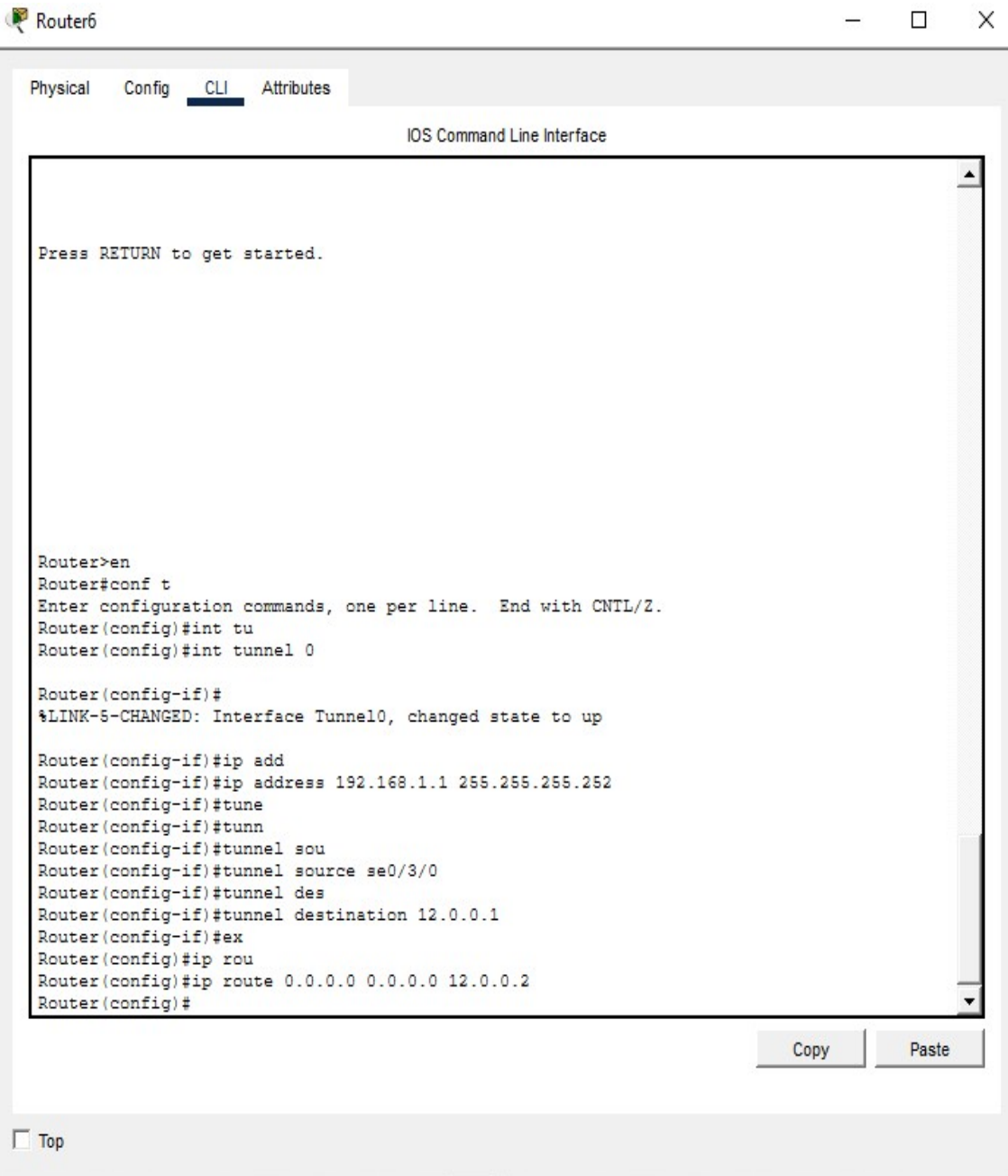
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, Serial0/3/0
L       11.0.0.1/32 is directly connected, Serial0/3/0
    13.0.0.0/24 is subnetted, 1 subnets
O       13.0.0.0/24 [110/1001] via 192.168.1.2, 00:00:21, Tunnel0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected, Tunnel0
L       192.168.1.1/32 is directly connected, Tunnel0
S*     0.0.0.0/0 [1/0] via 11.0.0.2

Router(config-if)#
```

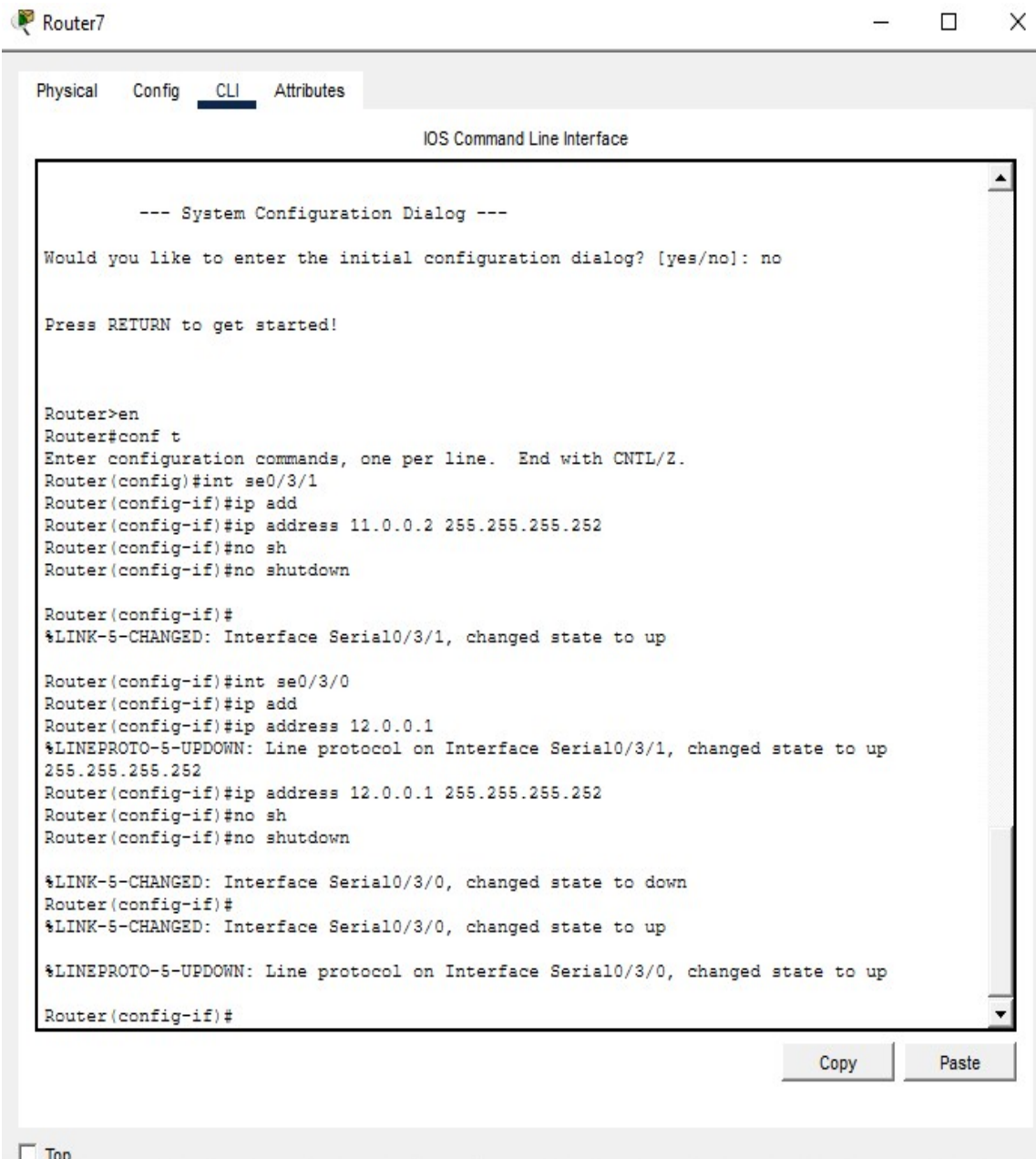
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R7:



R8:

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s0/3/1
Router(config-if)#ip add
Router(config-if)#ip address 12.0.0.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

Router(config-if)#int g0/0
Router(config-if)#ip add
Router(config-if)#ip address 1
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to
% Incomplete command.
Router(config-if)#ip add
Router(config-if)#ip address 13.0.0.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#|
```

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Router8

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
S* 0.0.0.0/0 [1/0] via 12.0.0.1

Router(config-router)#int se0/3/1
Router(config-if)#ip add
Router(config-if)#ip address 12.0.0.2 255.255.255.252
Router(config-if)#ex
Router(config)#router ospf 1
Router(config-router)#network 13.0.0.0 0.0.0.255 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#ex
Router(config)#
01:24:57: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Tunnel0 from LOADING to FULL,
Loading Done

Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 12.0.0.1 to network 0.0.0.0

    10.0.0.0/24 is subnetted, 1 subnets
O       10.0.0.0/24 [110/1001] via 192.168.1.1, 00:00:07, Tunnel0
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, Serial0/3/1
L       12.0.0.2/32 is directly connected, Serial0/3/1
    13.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       13.0.0.0/24 is directly connected, GigabitEthernet0/0
L       13.0.0.1/32 is directly connected, GigabitEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected, Tunnel0
L       192.168.1.2/32 is directly connected, Tunnel0
S* 0.0.0.0/0 [1/0] via 12.0.0.1

Router(config)#
```

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```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#tunnel 0
      ^
% Invalid input detected at '^' marker.

Router(config)#tunnel0
      ^
% Invalid input detected at '^' marker.

Router(config)#int tu
Router(config)#int tunnel 0

Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

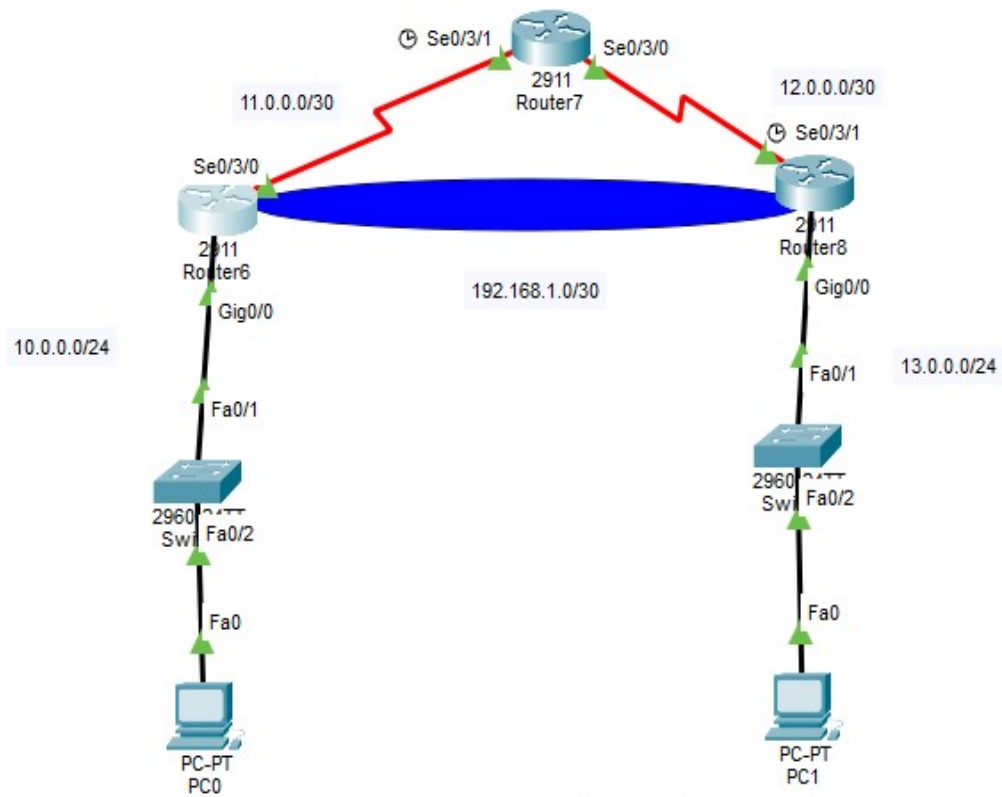
Router(config-if)#ip add
Router(config-if)#ip address 192.168.1.2 255.255.255.252
Router(config-if)#tunnel se0/3/1
      ^
% Invalid input detected at '^' marker.

Router(config-if)#tunnel source se0/3/1
Router(config-if)#tunnel des
Router(config-if)#tunnel destination 11.0.0.1
Router(config-if)#ex
Router(config)#ip route 0.0.0.0 0.0.0.0 12.0.0.2
%Invalid next hop address (it's this router)
Router(config)#ip route 0.0.0.0 0.0.0.0 12.0.0.1
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

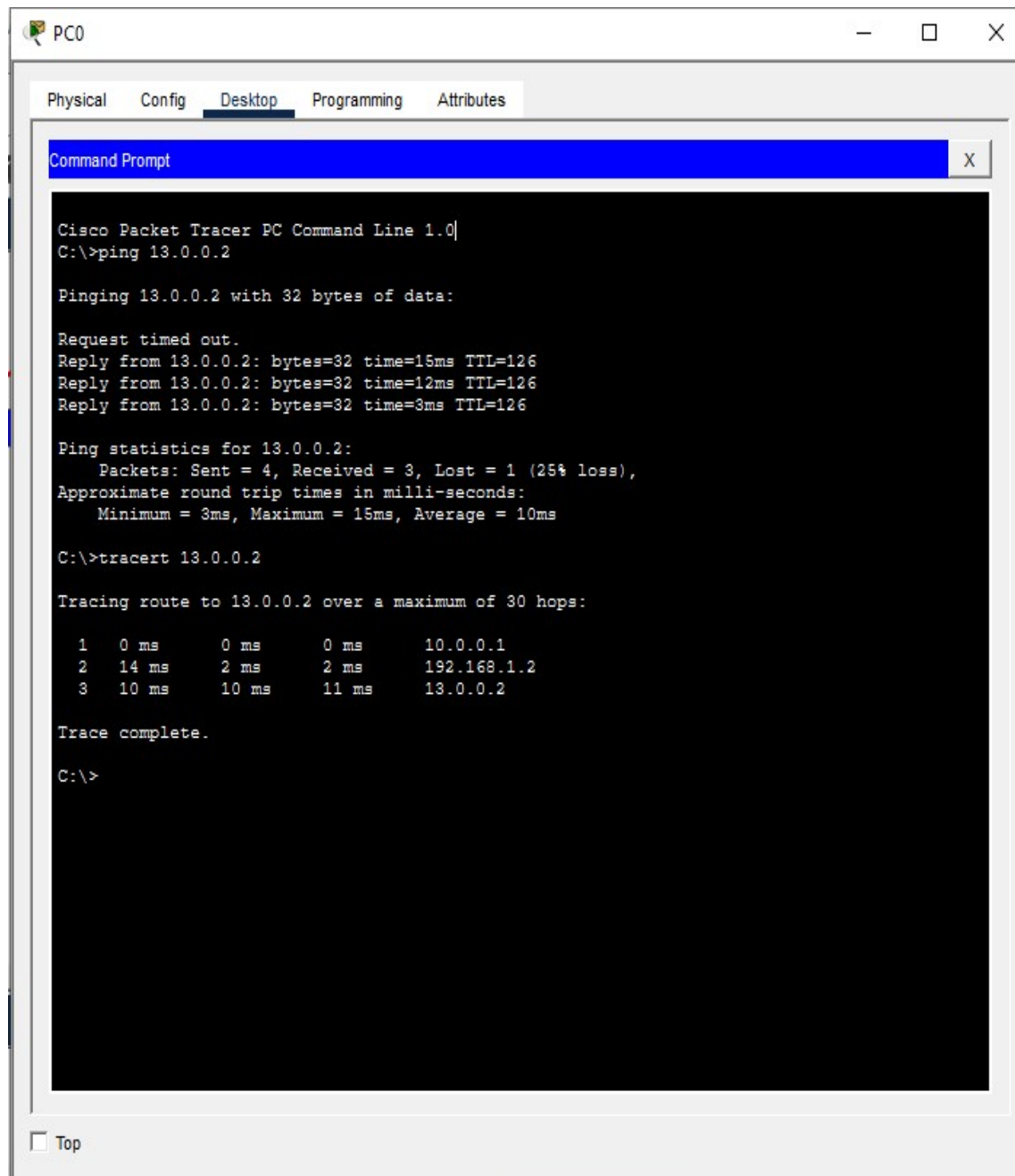
Router(config)#do sh
Router(config)#
```

☐ Top

Topology:



Testing:



8.5 Conclusions:

The Importance Of VPN is to make your connection secure , encrypted , we can't deny importance of vpn in our life in enterprises , companies as we show in out experiment.